Reviewer Pack — Additive Energy Threshold Exceeds ACC⁰ Circuit Size

Entry 916f234e492e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Entry ID: `916f234e492e` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-08 20:09:10 UTC

1. Conjecture

Field A (mathematical branch): Additive Combinatorics **Field B** (complexity object): ACC⁰ Circuit Size

Statement:

> For a Boolean function f: {0,1}^n → {0,1}, let E(f) be its additive energy: E(f) = |{(a,b,c,d) ∈ {0,1}^{4n} : f(a) + f(b) = f(c) + f(d)}|. Then, for all f with E(f) ≥ n^{2.5}, the minimal ACC⁰ circuit size computing f is ≥ n^{1.5} log n.

Rationale (proposer's reasoning):

> Additive energy captures structural regularity in Boolean functions, while ACC⁰ circuits lack depth to exploit such regularity. High additive energy implies complex interaction patterns that require more gates to simulate, exposing a separation between pseudorandom-like functions and shallow circuit classes.

Taxonomy category: `ACC_SIPSER` (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `384f79f3ae7826ee`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> default: >=80% seeds must support, no counterexample

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM_1	LLM_2

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=124, elapsed=240.3s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def add(a, b):
        return a + b

    def mod2(x):
        return x % 2

    def E(f):
        count = 0
        for a in range(1 << n):
            for b in range(1 << n):
                for c in range(1 << n):
                    for d in range(1 << n):
                        if add(mod2(f[a]), mod2(f[b])) == add(mod2(f[c]), mod2(f[d])):
                            count += 1
        return count

    def ACC0_circuit_size(n):
        # Simulate a depth-3, size 2^n ACC0 circuit using DPLL-style backtracking
        # This is a simplified version and may not be optimal for all functions
        def dpll(clauses, assignment):
            if not clauses:
                return True
            unit_clause = next((c for c in clauses if len(c) == 1), None)
            if unit_clause:
                literal = unit_clause[0]
                if literal < 0 and literal in assignment:
                    return False
                assignment[literal] = True
                new_clauses = [c for c in clauses if literal not in c and -literal not in c]
                if dpll(new_clauses, assignment):
                    return True
                del assignment[literal]
                assignment[-literal] = True
                new_clauses = [c for c in clauses if -literal not in c and literal not in c]
                if dpll(new_clauses, assignment):
```

```
                    return True
                del assignment[-literal]
                return False
            pure_literal = next((l for l in range(1, 2*n+1) if (l not in assignment and -l not in
            if pure_literal:
                assignment[pure_literal] = True
                new_clauses = [c for c in clauses if pure_literal not in c and -pure_literal not i
                if dpll(new_clauses, assignment):
                    return True
                del assignment[pure_literal]
                assignment[-pure_literal] = True
                new_clauses = [c for c in clauses if -pure_literal not in c and pure_literal not i
                if dpll(new_clauses, assignment):
                    return True
                del assignment[-pure_literal]
                return False
            return False

        n_vars = 2 * n
        clauses = []
        for i in range(1 << n):
            clause = [i + 1, -(i + 1)]
            clauses.append(clause)

        assignment = {}
        if dpll(clauses, assignment):
            return 2 ** n
        else:
            return float('inf')

    n = 40
    f = [random.randint(0, 1) for _ in range(1 << n)]
    E_f = E(f)
    circuit_size = ACC0_circuit_size(n)

    metric_name = "Additive Energy Threshold"
    metric_value = E_f
    instances_tested = 1
    conjecture_holds = E_f >= n ** 2.5 and circuit_size >= n ** 1.5 * math.log(n)
    counterexample = "" if conjecture_holds else f"Counterexample found: E(f)={E_f}, circuit_size=

    return {
        "metric_name": metric_name,
        "metric_value": metric_value,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30))  # Default to first 30 primes if

    results = []
    for seed in seeds:
        result = run_trial(seed)
```

```
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(result
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not r["conjecture_holds"] for r in results) and support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        counterexample_desc = "E(f) < n^2.5 or circuit_size < n^(1.5)*log(n)"
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample_desc}\" first_failing_seed={fir
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TIMEOUT after 240s
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> Test timed out before producing results, preventing validation of support fraction or counterexample detection. | next: Increase timeout duration and re-run test with extended computation limits

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	65999
2	preregistration	ollama_remote	qwen3:8b	0	0	26767
3	novelty	ollama_remote	qwen3:8b	0	0	20718
4	novelty	ollama_remote	qwen3:8b	0	0	16968
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17086
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12726
7	judge	ollama_remote	qwen3:8b	0	0	57138

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 217404 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in `research/audit/916f234e492e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/916f234e492e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/916f234e492e.tar.gz` (if generated)